

Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is the field of computer science concerned with creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, perception, language understanding, and decision making.

Machine Learning (ML) is a subfield of AI. Instead of explicitly programming every rule, a machine learning system learns patterns from data and uses those patterns to make predictions or decisions.

Common machine learning approaches include supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labeled examples, while unsupervised learning searches for patterns in data without predefined labels.

Applications of AI and machine learning include recommendation systems, fraud detection, medical image analysis, speech recognition, computer vision, and natural language processing.

A major challenge in AI systems is ensuring that models produce reliable results. Data quality, model evaluation, bias, privacy, and security are important considerations when deploying AI applications.